

ALLEN LU ZHAO QUAN

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EDUCATION

Nanyang Technological University (Data Science and Artificial Intelligence) **Aug 2023 – Dec 2027**
Bachelor of Science with Honours

- Cumulative GPA: 4.33 / 5.00 | Second Upper Class Honours (Distinction)
- Relevant Coursework: Data Analytics, Machine Learning, Databases, Calculus, Linear Algebra, Probability, Regression Analysis

WORK EXPERIENCE

PhillipCapital **Jan 2026 – Jul 2026**
Intern, Quantitative Risk

- Enhanced current GLM pricing model (Poisson/Gamma) to quantify rating factor relativities for private motor portfolios, exploring alternative models including Tweedie and GAM, Tweedie model achieved a Gini gain of 0.126, outperforming baseline model
- Designed Monte Carlo simulations to model stochastic claim outcomes for 9,253 motor insurance policies under a Tweedie GLM framework, identifying 99th-percentile loss ratio ~4.54% above expected levels, quantifying portfolio tail risk
- Developed SQL pipelines in Snowflake to compute borrower-level credit metrics (utilisation ratios, working-capital exposure, investment limits) and implemented rule-based classification to surface eligibility signals via Tableau dashboards for loan decisioning

Adroit Investment Technologies **May 2025 – Aug 2025**
Intern, Strategy & Business Intelligence

- Researched on emerging AI technologies across various sectors and prepared teasers, industry outlook, statistical reports for prospective clients, achieving 33% increase in client acquisition for the Month of May
- Managed components of financial statements using Power BI including Capital Accounts, Capital Calls, Capital Distribution
- Operated across a full stack of a Learning Management System (LMS) for a children's learning school using Odoo platform, including UI/UX adjustments, business logic workflows, data model configurations. Design shortlisted by CEO to be used as pitching material
- Implemented Excel algorithms to clean and compare 2,000 entries of investment data, enabling strategic report generation and stakeholder-ready insights

AIA **May 2024 – Jun 2024**
Intern, Actuary

- Devised key presentations for insurance companies, namely DirectAsia and Allied World Assurance regarding Customer Acquisition and Financial Modelling, resulting in a projected 20% increase in client count in coming 1-3 years, won overall best ideation
- Analysed option-based policy payoff structures (Call, Put, Straddle, Butterfly, Condor) to understand hedging mechanics in insurance product design

PROJECTS

SEA X Codex Hackathon 2026 (Sea Group X OpenAI) **Jun 2026**
Github Link: <https://github.com/Darkmatter126ms/Project-Naik>

- Designed a Tweedie GLM pricing kernel modelling expected weekly income loss for parametric micro-insurance, using BMKG flood risk scores and district-level rainfall triggers as rating factors across 50 synthetic gig worker profiles, achieving a Gini of 0.46 on holdout
- Deployed Tweedie GLM live within a four-agent pipeline with deterministic pricing. Team shortlisted for top 10% of 1,200+ applicants

Nanyang Fintech Catalyst FinTech Innovator's Hackathon 2026 **Mar 2026**
Github Link: <https://github.com/Darkmatter126ms/Wealth-Wellness-Hub>

- Built a research-calibrated, risk-adaptive financial wellness scoring model using CRRA utility and AHP-based factor weighting to quantify portfolio health across liquidity, diversification and growth, with profile-specific non-linear scoring derived from investor risk tolerance

Data Decomposition using Clustering Analysis **Aug 2025 – Nov 2025**
Github Link: <https://github.com/Darkmatter126ms/Data-Decomposition-using-Clustering-Analysis>

- Applied K-Means, K++-means, Agglomerative and Gaussian Mixed Models to decompose high-dimensional datasets, analysing cluster stability, probabilistic membership, and regime segmentation
- Evaluated model performance including Silhouette Score, Davies–Bouldin Index, Inertia, emphasising model assumptions and limitations

Morgan Stanley CodeToGive Asia Hackathon 2025 **Aug 2025**
Github Link: <https://github.com/Darkmatter126ms/CodeToGive>

- Enhanced a Donation dashboard to streamline donations, reduce donor attrition and improve outreach. Project shortlisted for overall Top 4
- Developed AI-powered donor badge generation by integrating SiliconFlow API, programmatically generating a custom badge in <4 seconds

EXTRA-CURRICULAR ACTIVITIES

Nanyang FinTech Catalyst (NFC) **Aug 2025 – Present**
Analyst-In-Training, Quantitative Research

- Spearheaded a team project on pricing strategies by developing a Beta Estimator via OLS regression and constructed a mean–variance portfolio optimiser. Risk-adjusted performance for Max–Sharpe strategies is benchmarked against naïve equal-weight allocation
- Backtested alpha signals on WorldQuant BRAIN using subindustry neutralisation, signal decay, and factor transformations on equity data. Best performing signal on adjusted net income achieved out-of-sample Sharpe of 2.04 with OS/IS ratio of 1.16 across TOP3000 US equities.

SKILLS & INTERESTS

Languages: English, Chinese (Native or Bilingual Proficiency), Italian (Limited Working Proficiency)

Skills: Python (libraries), R, SQL, C, C++, Java, JavaScript, HTML, CSS, PyTorch, Spark, Snowflake, Tableau, Bloomberg Terminal, Capital IQ, DCF Modelling, Options Pricing, Microsoft Office (Excel, PowerPoint, Power BI), VBA

Certifications: Solving Ordinary Differential Equations with MATLAB (MathWorks), Vector Calculus for Engineers (HKUST)

Interests: Gym, Volleyball, Basketball, Tennis, Poker & Mahjong, Probabilistic Thinking, Debate, Western Culinary, FX, Defi, Equities

Favourite Books: Thinking in Bets (Annie Duke), One Up on Wall Street (Peter Lynch)